Project Bill of Materials (BOM) - 270 Bolla Ave

Project Name: 270 Bolla Ave
Document Type: Integrated Systems Specification
Integrator Draft: v1.8 (Baseline Control & HVAC Integration)

Revision History

Version	Date	Description	Status
v1.0	2026-01-27	Baseline Release: Phased Infrastructure & Subsystem Segregation	Archived
v1.8	2026-01-27	Added Baseline HVAC Logic, Basalte Deseo UI, and verified URLs	**Current**

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1. Project Objectives & System Organization

The automation system at 270 Bolla Ave is designed for maximum reliability, security hardening, and high-fidelity environmental control. The architecture is organized into three localized **Load Control Panels (LCPs)** to minimize home-run copper and isolate failure domains:

- **LCP-1 (Garage)**: The "Power Hub" (12 Circuits). Central living areas, kitchen lighting, and high-inrush holiday circuits.
- **LCP-2 (Office 2)**: The "Environmental Hub" (11 Circuits). Master Wing, towel warmers, ventilation logic, and weather station.
- **LCP-3 (Tech Room)**: The "Security Island" (5 Circuits). 100% battery-backed, DC-powered enclosure for entry and 3D vision.

Core Standards: All lighting is **DALI-2** (addressable). All logic and switching is **KNX** (distributed). | :--- | :--- | :--- | :--- | | **Power Feeds**: | **LCP-1**: 12x 15A | **LCP-2**: 11x 15A | **LCP-3**: 5x 15A (+ Rack 30A) |

Section 1: High-Voltage (HV) Infrastructure

Main Service & Smart Panel Telemetry

External MPN	Manufacturer	Internal ID	Description	Qty	Phase	Notes
1-03294-02	SPAN	SPAN-GEN2	Smart Panel Gen 2 (32-Circuit)	2	1	Main Service (Local API)
MAIN-16	SPAN	SPAN-SUB	Smart Panel 16 (16-Circuit)	1	1	Sub-Panel (Logic/Tech)
-	Contractor Choice	SER-2/0-AL	#2/0 AWG Aluminum SER	-	1	Sub-SPAN Feed (100A)
-	Contractor Choice	NM-B-12/2	12/2 Romex (20A Branch)	-	1	Fridge, Dishwasher, Outlets

-	Contractor Choice	NM-B-6/3	6/3 Romex (50A Circuit)	-	1	Electric Range / Induction
-	Contractor Choice	NM-B-14/2	14/2 Romex (Standard)	-	1	General Lighting / LCP Feeds

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Section 2: Load Control Panel 1 (LCP-1) (Garage) - Kitchen/Main Hub

The automation core for the central living areas.

- **Enclosure**: Saginaw SCE-24H2408LP
- **Dimensions (Outside)**: 24.00" x 24.00" x 8.00" (610 x 610 x 203 mm)
- **Dimensions (Inside Usable)**: 24.00" x 24.00" x 7.75" (610 x 610 x 197 mm)
- **Depth Note**: 8" (203mm) nominal depth easily accommodates the 128.5mm (5.1") Mean Well SDR-480.
- **Internal Outlets**: Install 1x internal 15A duplex receptacle for plug-in logic (UniFi switch/spares).

External MPN	Manufacturer	Internal ID	Description	Qty	Phase	Notes
SCE-24H2408LP	Saginaw	SAG-24H2408	24x24x8 NEMA 4/12 Enclosure	1	1	Hinged Cover
SDR-480-24	Mean Well	MW-SDR-480	480W 24V DC PSU (UL 508)	1	1	Main Logic Power
SV/S 30.640.5.1	ABB	ABB-KNX-PS	KNX Power Supply 640mA	1	1	Powers Line 1.1
SV/S 30.160.1.1	ABB	ABB-PS-160	KNX Power Supply 160mA	1	1	Backbone Power
LK/S 4.2	ABB	ABB-LK-S	KNX Line Coupler	1	1	1.1 to Backbone
BE/S 16.20.3.2	ABB	ABB-BIN-16	16-fold Binary Input	1	1	Local Contacts
CBM E4 24DC/0.5-10A-NO	Phoenix Contact	PHX-CBM-E4	Electronic Circuit Protector	1	1	NEC Class 2
5WG1141-1AB03	Siemens	SIE-DALI-GW	KNX/DALI Gateway Twin N 141/03	1	1	Universes 1 & 2
USW-Pro-8-PoE	Ubiquiti	UI-SW-PRO-8	UniFi Pro 8 PoE (10G SFP+)	1	2	(Bernie) Garage Hub Switch
MDT-SCN-MBGW.01	MDT	MDT-SCN-MB	KNX/Modbus Gateway	1	2	Telemetry Bridge

1060/A	eldoLED	ELD-PWR-100	100W DALI-2 LED Driver	6 (TBD)	1	Area Tape Lights
86458668	Lunatone	LUN-DALI-010	DALI 0-10V PWM Interface	2	1	Guest / Half-Bath Vents
AKS-1216.03	MDT	MDT-AKS-1216	12-fold High-Inrush Actuator	1	1	Seasonal & WHF Power

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Section 3: Load Control Panel 2 (LCP-2) (Office 2) - Master Wing Hub

The automation core for the Master Bedroom, Bedrooms 3/4, and local environmental logic.

- **Enclosures**: nVent HOFFMAN ASE24X12X4NK / ASE12X12X4
- **Inside Dimensions**:
  - Top (24x12): 23.8" x 11.8" x 3.87" (605 x 300 x 98 mm)
  - Bottom (12x12): 11.8" x 11.8" x 3.87" (300 x 300 x 98 mm)
- **Depth Note**: **MAX COMPONENT DEPTH 3.5" (89mm)**. No SDR-series PSUs allowed here; use MDR or SV/S series only. (NUC fits @ 37mm).
- **Internal Outlets**: Install 1x internal 15A duplex receptacle for localized IT/logic gear.

External MPN	Manufacturer	Internal ID	Description	Qty	Phase	No
ASE24X12X4NK	nVent HOFFMAN	HOF-ASE24X12X4	Automation Enclosure (24x12x4)	1	1	Top
ASE12X12X4	nVent HOFFMAN	HOF-ASE12X12X4	Network/Aux Enclosure (12x12x4)	1	1	Bo
USW-Pro-8-PoE	Ubiquiti	UI-SW-PRO-8	UniFi Pro 8 PoE (10G SFP+)	1	1	(Be Off
NUC13ANKi5	ASUS	ASUS-NUC-13	NUC 13 Pro (i5, 16GB RAM)	1	1	(Be Log
SV/S 30.640.5.1	ABB	ABB-KNX-PS	KNX Power Supply 640mA	1	1	Pov
LK/S 4.2	ABB	ABB-LK-S	KNX Line Coupler	1	1	1.2
IPS/S 3.1.1	ABB	ABB-IPS-S	KNX IP Interface Secure	1	1	Log
USB/S 1.2	ABB	ABB-USB-S	KNX USB Interface	1	1	Loc
BE/S 16.20.3.2	ABB	ABB-BIN-16	16-fold Binary Input	1	1	Co

5WG1141-1AB03	Siemens	SIE-DALI-GW	KNX/DALI Gateway Twin	1	1	Un
INKNXCAR001I000	Intesis	INT-HVAC-CAR	Carrier/Day&Night to KNX Gateway	2	1	Ma
INKNXFGL001R000	Intesis	INT-HVAC-FUJ	Fujitsu to KNX Interface	1	1	Tec Sp
JAL-0810.02	MDT	MDT-JAL-0810	8-fold Shutter Actuator (24V DC)	1	1	Mo Sk
5WG1512-1CB01	Siemens	SIE-AKS-512	8-fold Load Switch (20A High-C)	1	1	To Ve
INKNXMBM100	Intesis	INT-MOD-KNX	HVAC Modbus to KNX Gateway	1	1	AC
SCN-MBGW.01	MDT	MDT-SCN-MB	KNX/Modbus Gateway	1	1	Ga Ter
MDR-100-24	Mean Well	MW-MDR-100-24	100W 24V DC DIN-Rail PSU	1	1	Lo
1060/A	eldoLED	ELD-PWR-100	100W DALI-2 LED Driver	6 (TBD)	1	Ma
86458668	Lunatone	LUN-DALI-010	DALI 0-10V PWM Interface	4	1	Ma / K
AKS-1216.03	MDT	MDT-AKS-1216	12-fold High-Inrush Actuator	1	1	To Att

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Section 3.1: Load Control Panel 3 (LCP-3) (Tech Room) - Door Entry MDF

The centralized "Security Island" - 100% DC-backed for maximum uptime.

- **Enclosure Option A (Standard)**: Saginaw SCE-24H2408LP (8" / 203mm Depth)
- **Enclosure Option B (Flush-Clean)**: Saginaw SCE-24H2406LP (6" / 152mm Depth)
- **Dimensions (Inside Usable A)**: 24.00" x 24.00" x 7.75" (610 x 610 x 197 mm)
- **Dimensions (Inside Usable B)**: 24.00" x 24.00" x 5.75" (610 x 610 x 146 mm)
- **Depth Rationale**:
  - **Bottleneck 1**: TP-Link Industrial Switch @ 120mm (4.7") depth.
  - **Bottleneck 2**: Yuasa 12Ah Batteries @ 98mm (3.86") depth.
  - **Wall Context**: A 2x6 wall (5.5" stud) + drywall (~6" total) makes Option B (6") nearly flush.

External MPN	Manufacturer	Internal ID	Description	Qty	Phase	Notes

SCE-24H2408LP	Saginaw	SAG-24H2408	24x24x8 NEMA Enclosure	1	1	Hinged Cover
eFlow104N	Altronix	ALT-EFLOW104N	10A Power Supply/Charger	1	1	Managed DC Rail
NP12-12	Yuasa	BATT-12AH	12V 12Ah SLA Battery	2	1	24V DC String
DDR-60G-15	Mean Well	MW-DDR-60G-15	24V to 15V DC-DC Converter	1	1	Power for NUC
NUC13ANKi5	ASUS	ASUS-NUC-13	NUC 13 Pro (i5, 16GB RAM)	1	1	(Bernie) Security Brain
IES210GPP	TP-Link	TPL-IES210GPP	Industrial 10-Port PoE+ Switch	1	1	(Bernie) DC-Input PoE Source
SR01	Akuvox	AKU-SR01	Secure Relay Module	4	1	Relay Board

3.2 Tech Room Power & Receptacle Spec

Differentiating Enclosure Power from Room/Wall Power

Load Category	Location	Receptacle	Circuit	Notes
Server UPS	Wall (Direct)	NEMA L6-30R	240V / 30A	Dedicated Twist-Lock
Workbench / Lab	Wall (General)	3x NEMA 5-20R	120V / 20A	General Bench Outlets
LCP-3 Logic	Inside Enclosure	Hardwired	120V / 15A	**Differentiated Circuit**
LCP-3 Service	Inside Enclosure	NEMA 5-15R	120V / 15A	For Switch / Comm.
General Room	Wall (Entry)	NEMA 5-15R	120V / 15A	Standard Code Circuit

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Section 4: Field Devices - KNX / Control

Sensors and User Interfaces (Not in Enclosures)

4.1 Sensors (Environmental & Presence)

| 056353 | Steinel | STE-TP-KNX | True Presence Multisensor KNX | 5 | 1 | Primary Bath + WC Cluster |

4.2 User Interfaces

External MPN	Manufacturer	Internal ID	Description	Qty	Phase	Notes

Sentido 4-way	Basalte	BAS-SENT-4	Sentido (4-Button) Switch	12	1	Native KNX
Deseo KNX	Basalte	BAS-DESEO	Basalte Deseo (OLED Thermostat)	1	1	Hallway Central UI
9011-02	Basalte	BAS-BUS-COU	KNX Bus Coupler for UI Units	13	1	For Sentido & Deseo

4.3 UI Integration Notes

- **Protocol**: Native KNX Multicast. All features (Buttons, RGB LED, Temp Sensor) are standard ETS Group Objects.
- **No Gateway**: Operates as a standalone bus node. Basalte Core server is NOT required for functionality.
- **Feedback**: Central RGB LED supports 1-byte or 3-byte status feedback for orientation or system alerts.
- **Mounting**: **CRITICAL**. Requires European round backbox pattern (60mm screw spacing). Do not use standard US rectangular mud-rings.

Section 5: Network Infrastructure (Core)

Standardized backbone for data connectivity.

External MPN	Manufacturer	Internal ID	Description	Qty	Phase	Notes
USW-Pro-Max-24-PoE	Ubiquiti	UI-SW-PRO-24	UniFi Pro Max 24 PoE	1	1	(Bernie) Core Network Switch
PDU15B10R	CyberPower	CYB-PDU-1U	1U Rack PDU	1	1	(Bernie) Rack Power
SMT1500C	APC	APC-UPS-1500	Smart-UPS 1500VA	1	1	(Bernie) Rack Backup

5.1 Wireless Infrastructure (WiFi 7)

External MPN	Manufacturer	Internal ID	Description	Qty	Phase	Notes
U7-Pro	Ubiquiti	UI-WAP-U7	UniFi7 Pro Access Point	4	1	(Bernie) High Reliability / WiFi 7
U-ACC-Pro-AP-Mount	Ubiquiti	UI-WAP-MOUNT	AP Pro Mounting Bracket	4	1	(Bernie) For ceiling tile/hard mount

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Section 6: Ventilation & Fans

Whole-House, Attic, and Bathroom Exhaust (EC Motors + DALI-2 Control)

6.1 Whole-House Fans (WHF)

External MPN	Manufacturer	Internal ID	Description	Qty	Phase	Notes
Stealth Pro 7.0X	QuietCool	QC-STL-7.0X	Stealth Pro 7.0X (ECM Motor)	2	1	Inspection-Critical
SR-2701S-DT7	Sunricher	SUN-2701S-DT7	DALI-2 Relay Puck	2	1	Power Control
SR-2401-10V	Sunricher	SUN-2401-10V	DALI to 0-10V Signal Converter	2	1	Speed Control

6.2 Attic Exhaust Fans

External MPN	Manufacturer	Internal ID	Description	Qty	Phase	Notes
AFG SMT PRO-3.0	QuietCool	QC-AFG-3.0	AFG SMT PRO-3.0 Gable Fan	4	1	Inspection-Critical
SR-2701S-DT7	Sunricher	SUN-2701S-DT7	DALI-2 Relay Puck	4	1	Power Control
SR-2401-10V	Sunricher	SUN-2401-10V	DALI to 0-10V Signal Converter	4	1	Speed Control

6.3 Bathroom & Laundry Exhaust

Location	Fan Motor	Silencer	Backdraft Damper	Diffuser (InviAir)
Primary Bath	Fantech FG 6M EC	Fantech LD 6	RSK 6	L100 (60") + S100 (6")
Bath 2 (Kids)	Fantech FG 6M EC	Fantech LD 6	RSK 6	L100 (48")
Bath 3 (Ens)	Fantech FG 6M EC	Fantech LD 6	RSK 6	L100 (48")
Guest Bath	Fantech FG 6M EC	Fantech LD 6	RSK 6	L100 (60")
Laundry Room	Fantech FG 6M EC	Fantech LD 6	RSK 6	S100 (8")

Half Bath	Fantech FG 6M EC	Fantech LD 6	RSK 6	S100 (6")

6.4 Environmental AI Inputs

External MPN	Manufacturer	Internal ID	Description	Qty	Phase	Notes
056353	Steinel	STE-TP-KNX	True Presence Multisensor KNX	5	1	Primary (2), Guest, Bath 3, Laundry
2064965	Warema	WAR-WETH-PRO	KNX Weather Station Pro	1	1	Environmental Hub
OAL-PM2.5	Temco	TEM-PM25	Outdoor PM2.5 Sensor (Modbus)	1	1	Wildfire SMK Sensor
SCN-MBGW.01	MDT	MDT-SCN-MB	KNX/Modbus Gateway	1	1	Protocol Bridge

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Section 7: Field Devices - Lighting (DALI-2)

Based on current floor plan placements.

External MPN	Manufacturer	Internal ID	Description	Qty	Type	Phase	Notes
X-Series	DMF Lighting	DMF-X2-SQ-FL	X-Series Square Flangeless	80	Recessed	1	Inspection-Critical
X-Series Wet	DMF Lighting	DMF-X2-WET	X-Series Sq Flangeless (Wet)	4	Recessed	1	Showers
Haiku 52"	Big Ass Fans	BAF-HAIKU-52	Haiku 52" Aluminum	3	Fan+Light	1	Essential Airflow
PCS-DUO	Southwire	CAB-NM-PCS	Romex PCS Duo (14/2 + 16/2)	3	Spools	1	Cabling Pulls

7.1 Decorative & Architectural Lighting (Phase-Ready TBD)

These loads are Phase 1 for rough-in/automation but fixtures are TBD. Pucks to be located in J-boxes.

External MPN	Manufacturer	Internal ID	Description	Qty	Phase	Notes

86458619	Lunatone	LUN-DALI-PD	DALI-2 Phase Dimmer (300W)	14	1	**FIXTURE TBD**: Decision Req.
TBD-DECO	Owner	DECO-MARKER	Fancy Floor-Plan Marker	14	1	For Planner Placement

- **Locations**: Front Entry (1), Foyer (1), Dining (1), Primary Bath (4), Other Baths (4), Bed 3, Bed 4, Guest Bed.
- **Note**: Primary WC is excluded from ornamental plan (Canned Lights only).
- **Requirement**: All decorative J-boxes must be 2-1/8" deep steel to house the DALI puck.

7.2 Outdoor Architectural Sconces (Phase 1 Ready)

High-performance perimeter lighting home-run to DALI-2 backbone.

External MPN	Manufacturer	Internal ID	Description	Qty	Phase	Notes
TBD-OUT-DALI	TBD (e.g. BEGA)	OUT-SCONCE	DALI-2 Native Outdoor Sconce	8	1	**FIXTURE TBD**: DALI-2 Native

- **Logic**: Must be specified as **DALI-2 Native** (with internal DALI driver).
- **Rationale**: Avoiding external DALI pucks in outdoor J-boxes minimizes moisture-failure points and simplifies IP65 sealing.
- **Requirement**: Decision Required in Phase 1 to ensure correct driver compatibility (Digital vs. Phase Cut).

Section 8: Physical Pathways (Conduit & Enclosures)

Item	Manufacturer	Description	Qty	Phase	Notes
BOX-SQ-4IN	Hubbell	4" Square Steel Box (2-1/8" D)	-	1	Rough-in Basic
BOX-FIRE-PAD	STI	SpecSeal Putty Pad	1	1	Fire Barrier
CON-EMT-1IN	Western Tube	1" EMT Conduit	2	1	Solar Roof
CON-EMT-2.5IN	Western Tube	2.5" EMT Conduit	2	1	Energy Wall

Section 9: Door Entry & Access Control (Akuvox)

3D Face Recognition & SIP Intercom System

External MPN	Manufacturer	Internal ID	Description	Qty	Phase	Notes

X915S	Akuvox	AKU-X915S	8" 3D Face Recognition Intercom	1	1	Main Entry
E16C	Akuvox	AKU-E16C	5" 3D Face Recognition Intercom	3	1	Secondary Entries
E16-A05H	Akuvox	AKU-E16-HOOD	Sun Shield / Rain Hood	1	1	See 'Accessories' tab
S567G	Akuvox	AKU-S567G	10" Android 12 Indoor Monitor	1	1	Internal Hub
1006-CS	HES	HES-1006	Heavy Duty Electric Strike	4	1	Lock Hardware
SR01	Akuvox	AKU-SR01	Secure Relay Module	4	1	Security Logic

Section 10: Environmental Sensors (Outdoor)

Field-mounted sensors home-run to LCP-2.

External MPN	Manufacturer	Internal ID	Description	Qty	Phase	Notes
2064965	Warema	WAR-WETH-PRO	KNX Weather Station Pro	1	1	North Wall
OAL-PM2.5	Temco	TEM-PM25	Outdoor PM2.5 Sensor (Modbus)	1	1	Smoke Interlock

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Section 11: Security & Tech Subsystem (Non-Critical)

Secondary compute, AI Pattern Learning, and Camera pre-wire.

External MPN	Manufacturer	Internal ID	Description	Qty	Phase	Notes
NUC13ANKi7	ASUS	ASUS-NUC-AI	NUC 13 Pro (i7, 32GB RAM)	1	2	(Bernie) AI Development Node (Owner)
MZ-V9P4T0B-AM	Samsung	SSD-990-4TB	990 Pro 4TB NVMe SSD	1	2	(Bernie) AI Storage

UVC-G5-Pro	Ubiquiti	UI-CAM-G5P	G5 Professional 4K Camera	12	2	(Bernie) **Phase 1: Pre-wire only**

Section 12: Routing & Pro-Integrator Notes

11.1 KNX Bus Topology (Redundant Loop Strategy)

- **Perimeter Run**: Route KNX Green Bus through all habitable walls at **48" AFF** (Switch Height).
- **Ceiling/Attic Spine**: Provide a central bus run through the attic with drops to all bathroom sensor locations.
- **Hardened Loop**: All major bus segments must be pulled as a **"Terminated Loop"**. Run the cable from the LCP, through the zone, and back to the LCP. **CRITICAL**: Only one end is to be connected to the bus power supply; the return tail must be labeled and capped.

11.2 LCP Service Spares & Pre-Wire

- **Window Shades (Pre-Wire)**: Pull **16/4 Shielded Cable** from LCP-1/2 to every window header. Terminate in "Router & Rescue" vaults per *window-shade-wiring-prep.md*. Coil 3ft of slack in LCP, label, and do not terminate.
- **LCP-1 (Garage)**: Pull **3x Spare Service Loops** (120V HV In + DUO Control Out) into the attic. **Note**: Requires an internal 15A outlet.
- **LCP-2 (Office 2)**: Pull **3x Spare Service Loops** (120V HV In + DUO Control Out) into the attic. **Environmental Hub**: Weather station and PM2.5 sensors terminate here. **Note**: Requires an internal 15A outlet.

11.3 Box & Finish Standards

- **Deep Boxes**: Standardize on 2-1/8" deep steel boxes to allow for DALI relay pucks and KNX pigtail service loops.
- **Floating Shields**: Drain wires on the KNX bus should be cut and taped at the device end; bond only at the LCP ground rail to prevent ground loops.
- **Internal Service Receptacles**: Every LCP (1, 2, and 3) must include an internal single or duplex 15A outlet mounted to the enclosure chassis for network switch adapters and service use.

11.4 Security Puck (SR01) Placement

- **Anti-Tamper Protocol**: All **Akuvox SR01** security relays must be centralized in **LCP-3** (Tech Room).
- **NO LOCAL LOGIC**: Under no circumstances should the relay be placed near the door terminal. The connection from door to panel is RS485 encrypted data only. This prevents "paperclip" attacks on the lock hardware.
- **Mounting**: All pucks to be DIN-rail mounted using official clips. Label each with the corresponding door name.

11.5 SPAN Panel Data Connectivity (AI Telemetry)

- **Physical Connection**: SPAN panels feature an **internal physical Ethernet port**. Use high-quality Shielded Cat6 (Cable Matters) for this link. Avoid Wi-Fi for telemetry to ensure zero-drop data for AI predictors.
- **Local API Gateway**: SPAN panels must be commissioned on the **Core LAN** (not a guest/IoT isolate) to allow the AI compute nodes to poll the local API.
- **Telemetry Frequency**: System must support high-frequency polling (~1Hz) of circuit-level power data for the AI pattern-learning predictors.

- **Token Persistence**: The "Door Proximity" authentication (3x button press) is a **one-time setup step** to generate a persistent `accessToken`. This token is intended for long-term integration and does **not** requires daily door cycling.
- **Handoff**: Ensure the integrator provides the permanent API token and static IP address for the Home Assistant SPAN integration bridge during site commissioning. 269: 270: ### **11.6 Energy Wall Phase Logic**

271: * **Phase 1 (Rough-in)**: Install all conduits (Section 8) and pull all 6x Cat6 Shielded cables (Section 18.2). This is critical to complete before drywall. 272: * **Phase 2/3 (Hardware)**: Installation of the Garage Hub Switch, Modbus Gateways, and Hybrid Inverters is deferred. LCP-1 must reserve 6 DIN-rail modules of space for future telemetry gear.

Section 12: Patio Infrastructure (Pre-Wire Only)

High-Current Heating & Outdoor Fan Control

External MPN	Manufacturer	Internal ID	Description	Qty	Phase	Notes
28829001	Southwire	CAB-10/2-NM	10/2 Romex (30A 240V)	3	1	Heater Pre-wire
PCS-DUO	Southwire	CAB-NM-PCS	Romex PCS Duo (14/2 + 16/2)	2	1	Fan Pre-wire
18/2 SHLD	Southwire	CAB-DALI-G	18/2 Shielded (DALI Bus)	1	1	Control Pre-wire

12.1 Patio Implementation Notes

- **Heaters (18kW Plan)**: Pull 3x 30A dedicated circuits. Control is via DALI-2 relay/contactor logic (hardware deferred).
- **Fans**: Pull power + control (Duo) for 2x Outdoor Haiku fans.
- **Surge Protection**: All outdoor data/bus lines must include a DIN-rail surge protector at the LCP entry point.

Section 13: Known Missing & Future-Phase TBDs

13.1 Aesthetic & Decorative Lighting (Pending Selection)

- **Chandeliers / Pendants**: Dining, Entry, and Kitchen Island fixtures are TBD. **Requirement**: All decorative junction boxes must be 2-1/8" deep to accommodate DALI-2 relay/dimmer pucks.
- **Vanity Sconces**: All bathroom vanity lighting locations and fixtures are TBD.
- **Outdoor Architectural Sconces**: Front entry and perimeter wall sconces are TBD.

13.2 Seasonal & Holiday Lighting (KNX Controlled)

- **Eave Outlets**: 2x Weatherproof outlets mounted under the roof eaves (North/South peaks) for Christmas/Holiday lighting. Home-run to **LCP-1**.
- **Ground Seasonal Outlets**: 2x Weatherproof outlets at ground level (Front/Side) for seasonal displays. Home-run to **LCP-1**.
- **Control Logic**: These 4 outlets must be terminated on a **KNX High-Inrush Relay Actuator** in LCP-1 for automated seasonal scheduling.

13.3 Exterior Infrastructure (Future Wiring)

- **Landscape Spine**: Pull 1" EMT conduit from Garage (LCP-1) to an external NEMA-3R landscape junction box for future landscape lighting expansion.
- **Under-Awning Future-Proofing**: Pull **14/2 Southwire Duo** to 4 corners of the exterior roofline for future architectural accent lighting.
- **Security Camera Drops**: Ensure all 12 Ubiquiti camera locations (Section 5) have Cat6 home-runs to the Tech Room (LCP-3).

13.4 Closet & Storage Automation (WIC / Linen / Pantry)

- **Reed Switches (Primary)**: Install recessed magnetic contact switches (e.g., Tane Pill-V) in all closet door frames. Home-run 18/2 to the nearest LCP Binary Input.
- **Stealth mmWave Pre-Wire**: Pull a KNX Bus "tail" to the center ceiling of each WIC and Linen closet. Coil 2ft of slack **above the drywall** (Stealth Vault).
- **Future sensor**: Prepared for **Creatrol 24G/60G mmWave** hidden installation (drywall penetration mode).
- **Requirement**: NO physical light switches are to be installed in closets or small storage areas. Lighting must be 100% sensor/contact driven.

Section 15: Bathroom Amenities (Towel Warmers)

Electronic climate comfort home-run to LCP relay actuators.

External MPN	Manufacturer	Internal ID	Description	Qty	Phase	Notes
[HREH06]()	Hudson Reed	REED-TW-LRG	2-Towel Electric Warmer (Chrome)	1	1	Large Chrome Collection
[HREH02]()	Hudson Reed	REED-TW-SML	1-Towel Electric Warmer (Chrome)	5	1	Small Chrome Collection

15.1 Towel Warmer Notes

- **Control**: Home-run to **LCP-1/2 High-Inrush Relays**. No wall switches.
- **Aesthetic**: Clean lines, polished chrome, hidden cable kit.
- **Capacity**: Primary Bath unit must handle 2 towels; all others 1 towel.

Section 14: Products Struggling / TBD Direct Links

The following products have generic or reseller links and require further direct manufacturer verification:

- **Akuvox E16-A05H Hood**: Currently linked to Low Voltage Dealer. Direct Akuvox accessory page not found.
- **Unitronic KNX Green Bus**: Current link is LAPP region-specific. Need a global spec sheet for North American compliance (LAPP USA).
- **Warema Weather Station Pro**: Product ID 2064965 confirmed, but technical documentation is region-locked.
- **MDT SCN-MBGW.01**: Verified as correct Modbus Gateway; need to confirm specific firmware features for AI PM2.5 interlock.

Section 16: Infrastructure - Unified Cabling Schedule

Standardized wire types for all subsystems.

Subsystem	Wire/Cable Type	Manufacturer	External MPN	Phase	Notes
Power (Core)	#2/0 AWG SER Aluminum	Contractor Choice	-	1	SPAN Panel Feeds (100A)
Power (Core)	12/2 Romex SIMpull	Contractor Choice	-	1	Fridge, DW, 20A Branch
Power (Core)	6/3 Romex SIMpull	Contractor Choice	-	1	Electric Range / Induction
Power (Core)	14/2 Romex SIMpull	Contractor Choice	-	1	Standard 15A Lighting/LCP
Lighting/Fans	Romex PCS Duo (14/2 + 16/2)	Southwire	PCS-DUO	1	Power + Control
KNX Bus	18/2 Shielded Twisted Pair	Belden	8760	1	US Equivalent for KNX Bus
DALI/Signal	18/2 Shielded Control	Southwire	18-2-SHLD	1	0-10V or DALI Bus
Network (SHLD)	Cat6 Shielded (F/UTP)	Cable Matters	160012	1	Core Network Standard
Network (Bulk)	Cat6 UTP (Bulk)	Cable Matters	160010	1	General Wall Jacks
Backbone (Fiber)	4-Core OS2 Singlemode	FS.com	151525	1	48-Fiber LC Patch Panel (OM4)
Consumables	KNX Push-Wire Connectors	Wago	243-211	1	Red/Black (Box 50)

Section 18: Inter-Zone Backbone Schedule

Trunk lines linking regions to the Server Rack. Fiber provides electrical isolation and 10G+ potential.

Trunk Link	Primary Medium	Redundant Medium	Phase	Notes
LCP-2 (Office) <--> Rack	4-Core OS2 Fiber	1x Cat6 Shielded	1	Logic Hub Trunk (10G Link)
LCP-1 (Garage) <--> Rack	2x Cat6 Shielded	-	1	Uplink (Aggregated Data)

Energy Wall <--> LCP-1	6x Cat6 Shielded	1x 18/2 SHLD	1	Local Comm (Telemetery Hub)
LCP-3 (Security) <--> Rack	1x Cat6 Patch	-	1	Local Tech Room Link
Media Center <--> Rack	4-Core OS2 Fiber	1x Cat6 Shielded	1	8K Future-Ready Media Trunk

18.1 Fiber Deployment Notes

- **Fiber Selection**: Use **Armored Pre-terminated Singlemode (OS2)**. These come with pulling eyes and a protective sleeve.
- **Cost**: ~100ft runs are roughly **$75 - $110**. It is cheaper than high-end shielded copper over length.
- **Transceivers**: Use standard **10G SFP+ Single Mode modules** (~$20/ea). You can swap these for 25G/40G modules in the future without re-pulling the glass.
- **Difficulty**: With "armored" versions, you can pull them like standard Romex. No special high-sensitivity handling required.

18.2 Energy Wall Communication (Solar/Battery)

- **Infrastructure**: Pull **6x Cat6 Shielded (F/UTP)** cables from the Energy Wall junction box directly to **LCP-1 (Garage)**.
- **Rationale**: LCP-1 acts as the local "Data Aggregator." By terminating here, we can use local **Modbus-to-Ethernet gates** (Section 2) to bridge serial inverter data onto the network before sending it up the 2x Cat6 backbone to the Server Rack/AI Center.
- **Point-to-Point**: Critical interop (Inverter <-> Battery BMS) remains local on the wall; these cables are for "Northbound" data collection and external control.
- **Usage Map (at LCP-1 Switch)**:
  - **Drop 1-2**: RS485 Modbus for Inverter data.
  - **Drop 3**: CAN-Bus monitoring.
  - **Drop 4**: Local Ethernet Web Gateway.
  - **Drop 5-6**: External SPAN Panel Ethernet ports.
- **Labeling**: Both ends must be clearly labeled "ENERGY-COMM-1" through "ENERGY-COMM-6".

Section 17: Load Control Panel Circuit Schedule

Defines the 120V Feeder lines from the SPAN panels to the automation enclosures.

17.1 LCP-1 Feed Breakdown (Garage / Main)

ID	Load Description	Rating	Notes
L1-01	**Automation Logic**	15A	ABB/Siemens/MeanWell PSUs
L1-02	**Kitchen/Pantry Lighting**	15A	DALI-2 Driver Hub
L1-03	**Family/Living/Entry Lighting**	15A	DALI-2 Driver Hub
L1-04	**Exterior/Landscape Lighting**	15A	Perimeter Sconces / Pre-wire
L1-05	**Interior Accent Lighting**	15A	eldoLED Tape Drivers (6)

L1-06	**Whole House Fan 1 (HV)**	15A	8.6A High-Inrush (EC Motor)
L1-07	**Whole House Fan 2 (HV)**	15A	8.6A High-Inrush (EC Motor)
L1-08	**Attic Fans / Ventilation**	15A	4x Gable Fans (4.0A Total)
L1-09	**Seasonal Outlets**	15A	KNX Relay (Eaves/Ground)
L1-10	**Reserved (Spare 1)**	15A	-
L1-11	**Reserved (Spare 2)**	15A	-
L1-12	**Reserved (Spare 3)**	15A	-

17.2 LCP-2 Feed Breakdown (Office / Master)

ID	Load Description	Rating	Notes
L2-01	**Automation Logic / NUC**	15A	Logic controller & Hub
L2-02	**Primary Suite Lighting**	15A	DALI-2 Zone
L2-03	**Bedrooms 2/3/4 & Hall Lighting**	15A	DALI-2 Zone
L2-04	**Primary/Master Accent Light**	15A	eldoLED Tape Drivers (6)
L2-05	**Primary Bath Towel Warmer**	15A	Dedicated KNX Relay
L2-06	**Guest/Kids Towel Warmer Group**	15A	~8A Continuous Load (5 rails)
L2-07	**Mechanical/Bath Vents**	15A	All Bedroom Wing Fans
L2-08	**Environmental (Skylight/Weath)**	15A	Logic + 24V Drive
L2-09	**Reserved (Spare 1)**	15A	-
L2-10	**Reserved (Spare 2)**	15A	-
L2-11	**Reserved (Spare 3)**	15A	-

17.3 LCP-3 Feed Breakdown (Tech Room / Security)

ID	Load Description	Rating	Notes
L3-01	**LCP-3 Logic (Managed)**	15A	Altronix EFlow 104N
L3-02	**Internal IT Receptacle**	15A	Switch / Aux / Service
L3-03	**Reserved (Spare 1)**	15A	-
L3-04	**Reserved (Spare 2)**	15A	-
L3-05	**Reserved (Spare 3)**	15A	-

Section 19: Architectural Fixture Decision Matrix

Decisions required in Phase 1 to ensure automation hardware (DALI-2) matches fixture electronics.

Area	Lighting Category	Hardware Needed	Phase	Decision Status
Outdoor Walls	Architectural Sconces	DALI-2 Native Driver	1	**TBD**: Need DALI-2 Native IP65
Front Entry	Foyer Pendant / Art	DALI-2 Phase Puck	1	**TBD**: Decision Req.
Dining Room	Large Feature Pendant	DALI-2 Phase Puck	1	**TBD**: Decision Req.
Primary Bath	Mirror / Ornamental (4)	DALI-2 Phase Puck	1	**TBD**: 4x Decisions Req.
Other Baths (4)	Aesthetic Vanities	DALI-2 Phase Puck	1	**TBD**: 1 per bath
Bedrooms (3)	Center Feature Lights	DALI-2 Phase Puck	1	**TBD**: Bed 3, 4, Guest
Main Garage	Workhorse LED Tape	DALI-2 LED Driver	2	**TBD**: Phase 2 Install
Living / Hall	Accent Pendants	DALI-2 Phase Puck	3	**TBD**: Phase 3 Finish

19.1 Decision Notes

- **DALI-2 Native vs. Puck**: Priority is to find DALI-2 native fixtures (especially for outdoor), but pucks (Section 7.1) are the fallback for standard 120V dimmable LEDs.
- **Box Depth**: Reminder for electrician—all fixture locations in this matrix **MUST** use 2-1/8" deep steel boxes to allow for recessed puck installation.